Diffie-Hellman

NU criptează!

Generează o cheie comună

Alice: $a = 11$

Bob: $b = 15$

Prim $p = 17$

$\alpha = 13$

$A = \alpha^a \bmod p = 13^{11} \bmod 17$

$= (-4)^{11} = -4 \cdot 4^{10} = -4 \cdot (4^2)^5 =$

$= -4 \cdot 16^5 = -4 \cdot (-1)^5 = -4 \cdot (-1) = 4.$

$B = \alpha^b \bmod p = 13^{15} \bmod 17$

$= (-4)^{15} \bmod 17 = 13^{11} \cdot 13^4 = 4 \cdot (-4)^4$

$= 4 \cdot 4^2 \cdot 4^2 = 4 \cdot (-1) \cdot (-1) = 4$

$K = B^a \bmod p = 4^{11} \bmod 17 = (4^2)^5 \cdot 4$

$= (-1)^5 \cdot 4 = -4 = 13$ cheia comună

$k = A^b \bmod p = 4^{15} \bmod 17$

$= 4^{11} \cdot 4^4 = 13 \cdot (4^2)^2 = (-4) \cdot \underbrace{(-1)^2}_{1}$

$k = -4 = 13$ ✓

Obs: $A^b = B^a$ MEREU!

El Gamal

Criptează!

I Generarea cheii de criptare

$G = \mathbb{Z}_{13}^*$ | $g = 5$?

Verific: $\langle g \rangle = G$?

x	1	2	3	4	5	6	7	8	9	10	11	12
5^x	5	12	8	1								1

$\Rightarrow 5^4 = 1 \Rightarrow \text{ord } 5 = 4$, $\langle 5 \rangle = \{1, 5, 8, 12\} \neq \mathbb{Z}_{13}^*$

$\Rightarrow$ 5 NU este generator

$g = 7$

x	1	2	3	4	5	6	7	8	9	10	11	12
7^x	7	10	5	9	(-2) 11	-14 -1 12	6	3	8	4	2	1

$13 \cdot 5 = 65$

$\Rightarrow \langle 7 \rangle = \mathbb{Z}_{13}^* \Rightarrow 7$ generator ✓

Lagrange: $\text{ord}(x \in G) \mid \text{ord}\, G$

$\text{ord}\, \mathbb{Z}_{13}^* = 12 \Rightarrow \text{ord}(x \in \mathbb{Z}_{13}^*) \in D_{12}$

Deci: $G = \mathbb{Z}_{13}^*$, $g = 7$, $q = 13$ $(q-1 = 12)$

Alg $x \in \{1, 2, \dots, q-1\} = \{1, 2, \dots, 12\}$

$x = 5$

Calculez: $h = g^x \bmod q = 7^5 \bmod 13 = 11$

"Criptarea"

Mesaj: $M = 8$

Def. $f: \mathbb{Z} \to \mathbb{Z}_{13}^*$, $f(x) = 2x - 1$

$f^{-1}(x) = \frac{x+1}{2}$

$M \longmapsto m = f(M) = 2 \cdot 8 - 1 = 15 \in \mathbb{Z}_{13}^*$

$= 2$

$M = 8$, $m = 2$

Alg $y \in \{1, 2, \dots, 12\}$: $y = 6$

Calculez : $S = h^y \bmod q = 11^6 \bmod 13$

$= (-2)^6 = 2^6 = 2^4 \cdot 2^2 = 3 \cdot 4 = 12.$

$c_1 = g^y \bmod q = 7^6 \bmod 13 = 12$

$c_2 = m \cdot S = 2 \cdot 12 = 24 \bmod 13 = 11.$

$M \xrightarrow{f} m \xrightarrow{\text{El G.}} \underline{(c_1, c_2) \text{ codul}}$

III Decriptarea

$c_1^x \bmod q = 12^5 \bmod 13 = (-1)^5 = -1 = 12 = S$

$S^{-1} = 12^{-1} = (-1)^{-1} = -1 = 12$

$m = c_2 \cdot S^{-1} = 11 \cdot 12 = (-2) \cdot (-1) = 2$

$M = f^{-1}(m) = f^{-1}(2) = \frac{3}{2} \neq M$ ✗

$3 \cdot 2^{-1} = 3 \cdot 7 = 21$

Rabin

Ian $p = 7$ $\quad 4k+3$

$q = 11$

$n = p \cdot q = 77$

$m \in \{0, \dots, n-1\} \Rightarrow \{0, \dots, 76\}$

Aleg $m = 23$

Criptarea: $c = m^2 \bmod n = 23^2 \bmod 77$

$= 67$

Decriptarea

$m_p = 67^{\frac{7+1}{4}} = 67^2 \bmod 7 = 4^2 \bmod 7$

$m_p = 2$

$m_q = 67^{\frac{11+1}{4}} = 67^3 \bmod 11 = 1 = m_q$

$\Rightarrow \pm m_p + p\mathbb{Z} = \pm 2 \text{ in } \mathbb{Z}_p = 2 \text{ sau } -2 \text{ in } \mathbb{Z}_7$

$\Rightarrow 2$ sau $5 \in \mathbb{Z}_7$ cele 2 rădăcini

$c = 67 \in \mathbb{Z}_7$

$\Rightarrow \pm 1$ sunt cele 2 rădăcini ale lui $c = 67$

$\in \mathbb{Z}_9 = \mathbb{Z}_{11}$

$(7, 11) = 1 \quad y_p \cdot 7 + y_q \cdot 11 = 1$ → $7y_p = 1 - 11y_q$

↑ Incerc: $y_p = -3$

$y_q = 2$

$r_1 = (y_p \cdot p \cdot m_q + y_q \cdot q \cdot m_p) \bmod n$

$= (-3 \cdot 7 \cdot 1 + 2 \cdot 11 \cdot 2) \bmod 77$

$= -21 + 44 = \boxed{23} \bmod 77$

$r_2 = (-3 \cdot 7 \cdot 10 + 2 \cdot 11 \cdot 5) \bmod 77$

$= -210 + 110 = -100 = -77 - 23$

$= -23 = 54$

$S_1 = (-3 \cdot 7 \cdot 1 - 2 \cdot 11 \cdot 2) \bmod 77$

$= -21 - 44 = -65 = 12$

$S_2 = -3 \cdot 7 \cdot 10 - 2 \cdot 11 \cdot 5 \pmod{77}$

$= -210 - 110 = -320 = -308 - 12$

$= -12 = 65$

Euclid extins:

$(13, 19) = 1$

$\exists \alpha, \beta$ a.î. $13\alpha + 19\beta = 1$

$a = 13$, $b = 19$, $a \neq b$ $\begin{cases} a > b \Rightarrow a \to a - b \\ a < b \Rightarrow b \to b - a \end{cases}$

$a_0 = a_1$, $b_0 = 19$ $\to$ $a_1 = 13$, $b_1 = 6$ ($b \to b - a$) $\to$ $a_2 = 7$, $b_1 = 6$ ($a \to a - b$) $\to$ $a_3 = 1$, $b_1 = 6$ ($a = a - b$) $\to$ $b \to b - a$ de 5 ori

$a = 1$, $b = 1$

$1 = b_1 - 5a_3 = b_1 - 5(a_2 - b_1)$

$= b_1 - 5(a_1 - b_1 - b_1) = (b_0 - a_0) - 5(a_0 - 2(b_0 - a_0))$

$= b_0 - a_0 - 5a_0 + 10(b_0 - a_0)$

$= b_0 - 6a_0 + 10b_0 - 10a_0 = 11b_0 - 16a_0$

$1 = \underset{\beta}{11} \cdot 19 - \underset{\alpha}{16} \cdot 13$